

# Matched-Replay Supplement for Conditional Routing for 2K Violence Detection: Accuracy, Invocation Cost, and Cross-Device Evaluation

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Table S3: Process and device telemetry for the matched replay. Values are mean  $\pm$  sample SD across three independent processes. Process wall time includes initialization and the 60-s source-time warm-up. Analysis-call latency covers RGB conversion and runtime processing in the 600-s measured interval but excludes source decoding and queuing. CPU follows `psutil` process semantics and may exceed 100%; RSS is process peak resident memory. GPU values are NVIDIA device telemetry.

| Workload     | Schedule             | Wall (s)         | Lat. p50        | Lat. p95        | Lat. p99        | CPU (%)           | RSS (GiB)       | GPU (%)       | VRAM (MiB)    | Power (W)      |
|--------------|----------------------|------------------|-----------------|-----------------|-----------------|-------------------|-----------------|---------------|---------------|----------------|
| Normal       | Conditional routing  | 426.5 $\pm$ 0.6  | 21.4 $\pm$ 0.1  | 299.4 $\pm$ 2.4 | 626.2 $\pm$ 4.7 | 571.4 $\pm$ 8.2   | 3.82 $\pm$ 0.14 | 9.9 $\pm$ 0.8 | 8042 $\pm$ 2  | 68.7 $\pm$ 0.5 |
|              | Continuous reference | 1860.7 $\pm$ 8.7 | 310.3 $\pm$ 2.3 | 383.1 $\pm$ 2.5 | 401.2 $\pm$ 2.7 | 1060.9 $\pm$ 11.8 | 2.05 $\pm$ 0.02 | 6.5 $\pm$ 1.0 | 7990 $\pm$ 13 | 66.8 $\pm$ 0.3 |
| Mixed        | Conditional routing  | 436.5 $\pm$ 0.6  | 21.4 $\pm$ 0.0  | 313.9 $\pm$ 3.2 | 639.2 $\pm$ 6.5 | 575.7 $\pm$ 9.8   | 4.41 $\pm$ 0.26 | 9.8 $\pm$ 0.2 | 8050 $\pm$ 15 | 70.2 $\pm$ 0.8 |
|              | Continuous reference | 1853.4 $\pm$ 3.1 | 308.3 $\pm$ 0.7 | 382.7 $\pm$ 1.5 | 400.3 $\pm$ 1.3 | 1066.4 $\pm$ 11.4 | 2.05 $\pm$ 0.01 | 7.1 $\pm$ 1.7 | 7990 $\pm$ 13 | 67.1 $\pm$ 0.6 |
| Kinetic-rich | Conditional routing  | 514.2 $\pm$ 1.0  | 23.5 $\pm$ 0.2  | 364.1 $\pm$ 0.7 | 662.5 $\pm$ 2.4 | 628.8 $\pm$ 10.3  | 4.07 $\pm$ 0.13 | 9.1 $\pm$ 1.2 | 8050 $\pm$ 14 | 71.1 $\pm$ 0.9 |
|              | Continuous reference | 1850.1 $\pm$ 8.1 | 307.3 $\pm$ 1.5 | 381.7 $\pm$ 3.0 | 399.8 $\pm$ 2.8 | 1069.2 $\pm$ 13.7 | 2.03 $\pm$ 0.00 | 7.1 $\pm$ 0.9 | 7989 $\pm$ 14 | 67.1 $\pm$ 1.0 |

Table S4: Per-update stage timing from the measured replay traces (ms; mean  $\pm$  sample SD across process means). YOLO/tracker is the complete detector-tracker call, while YOLO inference is the detector’s reported inference subset and must not be added to it. Classifier latency is summarized over all MoViNet calls in the same process, including source-time warm-up; the runner did not retain every per-call latency separately in the measured interval. Routed updates may emit more than one call; therefore the columns are not an additive latency decomposition.

| Workload     | Schedule             | Total             | YOLO/tracker     | YOLO infer.     | HDBSCAN         | Gate            | Classifier p50   | Classifier p95    |
|--------------|----------------------|-------------------|------------------|-----------------|-----------------|-----------------|------------------|-------------------|
| Normal       | Conditional routing  | 54.44 $\pm$ 0.08  | 12.37 $\pm$ 0.08 | 5.82 $\pm$ 0.06 | 3.75 $\pm$ 0.02 | 0.04 $\pm$ 0.00 | 14.35 $\pm$ 0.08 | 104.02 $\pm$ 3.95 |
|              | Continuous reference | 318.33 $\pm$ 1.68 | 0.00 $\pm$ 0.00  | 0.00 $\pm$ 0.00 | 0.00 $\pm$ 0.00 | 0.00 $\pm$ 0.00 | 14.15 $\pm$ 0.08 | 84.61 $\pm$ 2.34  |
| Mixed        | Conditional routing  | 56.23 $\pm$ 0.13  | 12.25 $\pm$ 0.02 | 5.75 $\pm$ 0.01 | 3.88 $\pm$ 0.01 | 0.04 $\pm$ 0.00 | 14.36 $\pm$ 0.06 | 103.33 $\pm$ 1.86 |
|              | Continuous reference | 316.88 $\pm$ 0.56 | 0.00 $\pm$ 0.00  | 0.00 $\pm$ 0.00 | 0.00 $\pm$ 0.00 | 0.00 $\pm$ 0.00 | 14.10 $\pm$ 0.01 | 85.76 $\pm$ 1.56  |
| Kinetic-rich | Conditional routing  | 69.23 $\pm$ 0.14  | 12.24 $\pm$ 0.07 | 5.71 $\pm$ 0.05 | 5.13 $\pm$ 0.01 | 0.05 $\pm$ 0.00 | 14.35 $\pm$ 0.09 | 100.92 $\pm$ 2.18 |
|              | Continuous reference | 316.19 $\pm$ 1.51 | 0.00 $\pm$ 0.00  | 0.00 $\pm$ 0.00 | 0.00 $\pm$ 0.00 | 0.00 $\pm$ 0.00 | 13.98 $\pm$ 0.29 | 85.97 $\pm$ 1.47  |